

Sean Balbale

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Available for full-time employment starting June 2027

SUMMARY

ECE/CS student building production-grade agentic AI systems: an LLM coaching agent orchestrated over a self-hosted RAG pipeline, deployed and operated on infrastructure I run myself. Alongside that, internship experience owning backend features end-to-end and debugging live production incidents at Sonos and Bullhorn. Comfortable working directly in ambiguous, real-world systems rather than notebooks.

SELECTED PROJECTS — MORE ON PORTFOLIO WEBSITE

Astraphe AI — Agentic AI Athletic Coaching Platform

2026

- Designed and shipped a production agentic coaching system end-to-end: a Gemma 4 LLM agent orchestrated over a pgvector RAG pipeline that retrieves structured coaching memory (training history, readiness, athlete goals) to ground its guidance, rather than a stateless chatbot wrapper.
- Built the orchestration and data layer in FastAPI: a multi-source ingestion pipeline (Strava, WHOOP, Garmin, intervals.icu) deduplicated onto canonical records, feeding a training-load engine (TSS, CTL/ATL/TSB, Strain) that grounds agent outputs in real athlete state.
- Self-hosted the full system in production on Kubernetes (k3s) with GitHub Actions CI/CD (80% coverage gate); owned the Svelte 5 / Capacitor mobile client (iOS/Android) end-to-end. Later added a standalone MCP server exposing user data to Claude via OAuth 2.1 — a second, independent tool-use integration; open-sourced under Apache-2.0.

Cherrybrook Homelab & HPC Cluster — Self-Hosted Infrastructure

2025 – Present

- Built and operate a four-node Proxmox VE cluster hosting Dockerized production workloads including Astraphe AI; executed a live VLAN/subnet migration with split-horizon DNS. Separately, assembled a three-node Slurm/MPI HPC cluster for parallel workload benchmarking.

WORK EXPERIENCE

Cloud Service Software Engineering Intern — Sonos | Boston, MA

Summer 2026

- Owned migration of two production Java/Spring Boot services from WebSocket command delivery to an SQS-backed queue end-to-end: designed the feature-flagged per-player fan-out with retry-until-acceptance semantics, and validated both transports against real speakers.
- Diagnosed a silent delivery failure in pre-production by tracing commands live through the SDK and queue infrastructure to an endpoint-routing defect, and drove the fix across both services.
- Ported the test suite's DynamoDB-stream Lambda into a local harness for pre-push validation; built Datadog dashboards for the new delivery path.

Software Engineering Intern — Bullhorn | Boston, MA

Summer 2025

- Owned the BH Boot 1.0.0 framework migration end-to-end: upgraded multiple production Java services from Spring Boot 2 to Spring Boot 3, refactoring all `javax` namespaces to `jakarta` and restructuring the proprietary data access layer.
- Led the Log4j2 security upgrade across SEARCH_AI, ATS_CRM, and MIDDLE_OFFICE service groups; refactored legacy TestNG/JMockit suites to JUnit 5 and Mockito.

Engineering Teaching Assistant — ENGR 212L Linear Circuit Theory, Trinity College | Hartford, CT

Spring 2026

- Led weekly lab sessions for ENGR 212L covering RLC circuit analysis, op-amps, Laplace transforms, Fourier series, and active filter design.

EDUCATION

Trinity College — Hartford, CT

Expected May 2027

Bachelor of Science in Electrical and Computer Engineering

GPA: 3.70

Bachelor of Science in Computer Science

Coursework: Deep Learning, Software Engineering, High-Performance Computing, Data Structures & Algorithms.

Extracurriculars: Varsity Athlete — Trinity College Men's Rowing; Singer & Technology Chair — The Trinity College Accidentals.

SKILLS

AI/LLM: LLM Orchestration & Agent Design, RAG (pgvector), MCP, Prompt Engineering, Gemma, OAuth 2.1

Languages: Python, Java, TypeScript, JavaScript, C, C++, SQL, MATLAB, LaTeX

Backend: FastAPI, Spring Boot 3, Jakarta EE, Node.js, REST APIs, JUnit 5, Mockito

Frontend: React, Next.js, Svelte, Tailwind CSS

Cloud & DevOps: Kubernetes, Docker, AWS, CI/CD (GitHub Actions), Proxmox, Git, Linux, Supabase, Slurm, MPI

Embedded / Hardware: Arduino, ARM, 8051, FPGA, BLE 5, SPI/I2C

RESEARCH & HONORS

“Embodied Harms and Inferred Data: Redefining Privacy in Extended Reality” — *Int'l Journal of Law, Ethics, and Technology* (in production); awarded the Alumni Prize in English Composition, Trinity College, May 2026.

PERSONAL

Serious home cook across a wide range of cuisines. Outside the kitchen: cycling, ski mountaineering, and tinkering on whatever project has captured my attention.